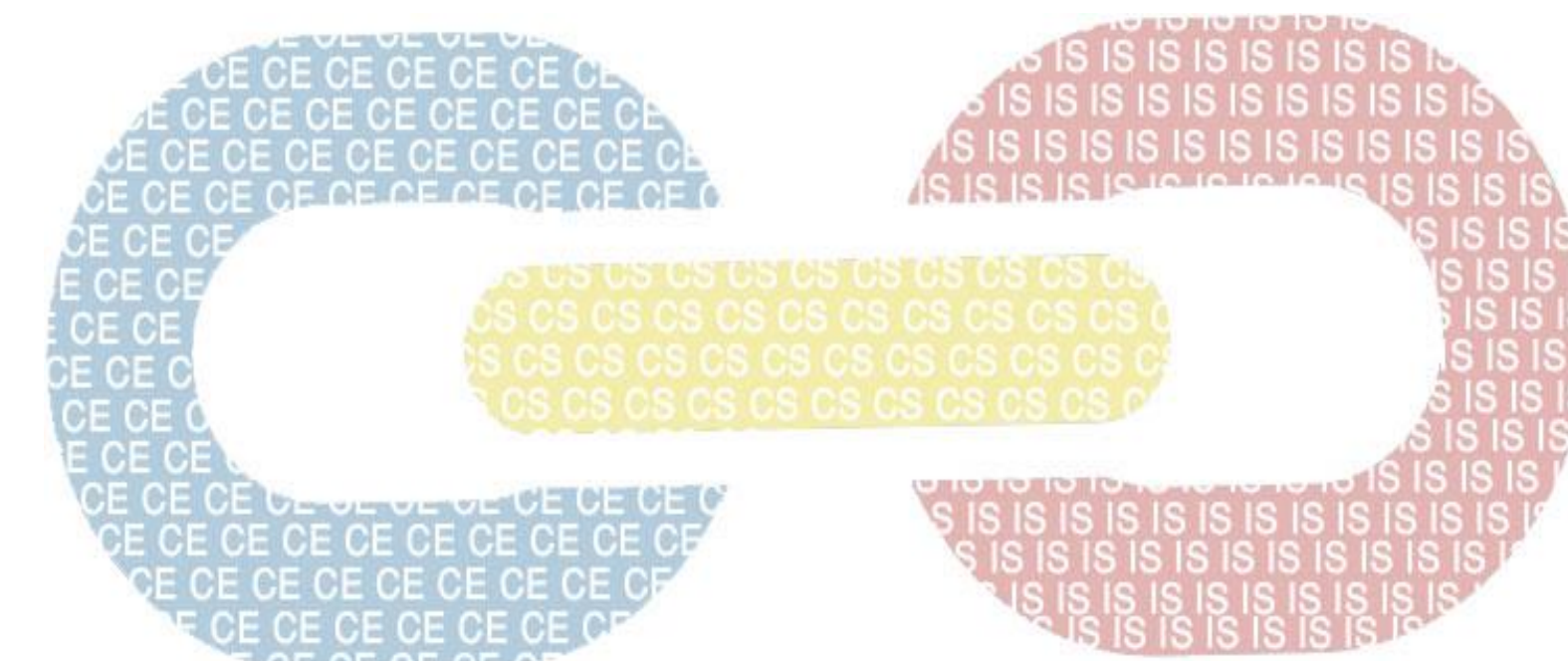




Secure Natural Language to Bash Execution - ShellSentry

Aljazi Ali Almujaaddam – 202202227 | Dana Hussain Alhayki – 20223431
Supervised By : Dr. Abdulla Khalifa Aldoseri

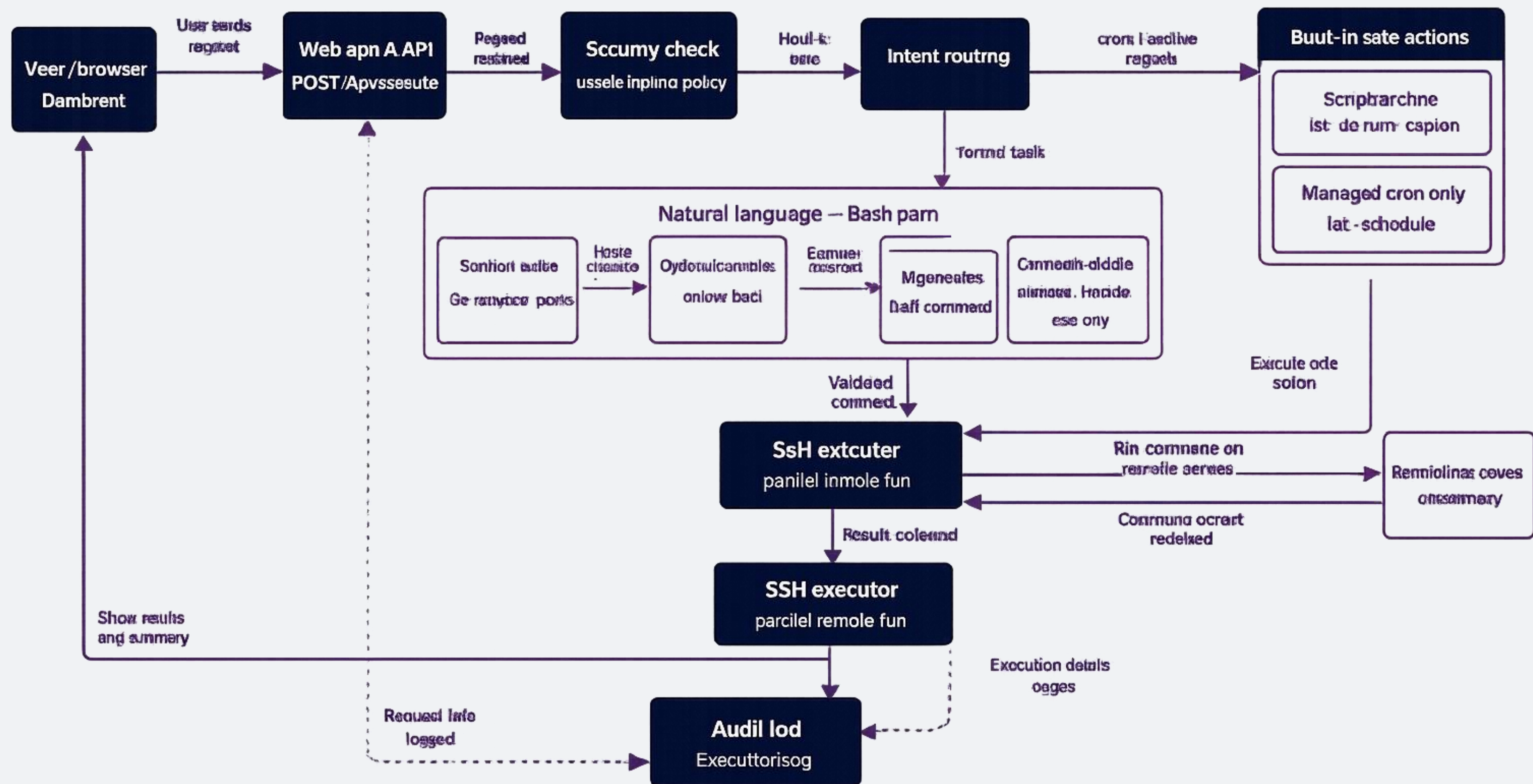
ABSTRACT

A secure, web-based prototype that takes admin requests in natural language to a validated bash command to execute on a remote Linux system. The project tackles the challenges of manual shell management and the execution of commands generated by AI, using a combination of authenticated access, retrieval-augmented generation (RAG), host-aware context collection, deterministic command validation, and SSH-based multi-server orchestration. Flask, Python, Paramiko, FAISS, and an OpenAI-compatible Large Language Model were used to implement the system. Functional, security, and performance testing are done in a controlled environment in the virtual lab. The results demonstrated successful authentication of execution, multi-host command orchestration, blocking of unsafe commands, and structured audit logging with a reliable report. ShellSentry showcases how AI-driven shell automation can become much more secure by implementing multiple-level validation, execution policies, and accountability.

OBJECTIVES

- Develop a secure web-based system that converts natural-language requests into Bash commands.
- Enable authenticated remote execution across multiple Linux servers via SSH.
- Reduce unsafe AI-generated command risks through deterministic validation and security controls.
- Improve command accuracy using host-aware context and retrieval-augmented generation (RAG).
- Provide audit logging, script management, and explainable execution reporting.

METHODS/DIAGRAMS/FIGURES



RESULTS

CONCLUSION & FUTURE WORK

REAL
RESULT
WORKFLOW

1. INPUT (NATURAL LANGUAGE REQUEST)

2. GENERATED BASH COMMAND

3. EXECUTION RESULTS (PER SERVER)

SERVER	STATUS	STDOUT (SUMMARY)
Server 1 (Ubuntu)	Success	Disk usage information retrieved successfully.
Server 2 (Kali Linux)	Success	Disk usage information retrieved successfully.
Server 7 (Raven Linux)	Success	Disk usage information retrieved successfully.

4. AI SUMMARY

☒ Disk usage collected successfully from all target servers.

- Conclusion
Secure AI-assisted Linux administration prototype
Natural language to validated Bash execution
Multi-server SSH orchestration with audit logging
Successfully blocked unsafe commands through layered security
- Future Work
Role-based access control (RBAC)
Human approval for sensitive commands
SIEM-integrated immutable audit logging
Production-ready deployment enhancements